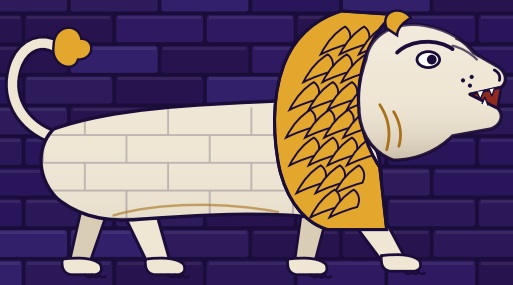
PHYSICS & ENERGY TECHNOLOGY · LEVEL 3

EM

ELECTRICITY AND MAGNETISM

A handout of ideas, facts and problems



12 sections · Edition of November 2026 · 40 pages

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How to read this handout

This handout follows the method of Jaan Kalda's physics handouts. The text is short; the work is in the numbered items. Read each section once, then return to its items until you can use them without looking.

Idea 1. An *idea* is a way of thinking that unlocks a whole family of problems. Ideas are framed in violet.

Fact 1. A *fact* is a law or relationship you are expected to know, with the meaning of every symbol and the conditions under which it holds.

Method 1. A *method* is a fixed sequence of steps for a standard task.

Pr 1. Problems are numbered through the handout. A problem followed by a *Solution* is worked for you; cover the solution and try first. For the remaining problems, attempt the problem, then use the *Hints* section, and only then the *Answers* section, both at the back of the subject.

Physical quantities are set in italic, units in roman, with a thin space between number and unit. The index at the end lists every topic and symbol with the page on which it is introduced.

Electricity and Magnetism

Concept-first coverage of the course plan. Chapters 23-31 have lecture support; inductance, AC and EM waves are course-plan extensions without separate lecture files in this folder. Advanced field integrals are recognition topics, not worked derivations.

1 Charges, conductors and Coulomb force

Aims. Predict attraction or repulsion from charge signs; Use simple Coulomb-force ratios without unnecessary arithmetic.
Before this section: Positive and negative charge; squares; micro means one millionth.

Charge is conserved and occurs in integer multiples of the elementary charge. Ordinary charging transfers electrons. Like charges repel and unlike charges attract. Conductors let charge move readily; insulators restrict it. A neutral object can still be attracted by polarization.

Idea 1. Coulomb-force ratios let you compare two situations without knowing every charge value. Imagine moving the same two charges farther apart: the effect spreads according to the square of separation. Charge signs tell attraction or repulsion, while charge sizes and separation set the magnitude.

Fact 1.

$$F = k|q_1 q_2|/r^2$$
$$k = 8.99 \times 10^9 \ \mathrm{N\,m^2/C^2}$$

where F is the electrostatic force magnitude (N); q_1, q_2 is the source charges (C); r is the separation of charges (m).
Conditions. Coulomb calculations here use point charges, or a stated model equivalent to point charges.

Why it holds. Doubling separation squares the distance factor, making the force one quarter as large. Changing only one charge magnitude scales the force directly without changing the separation factor.

Method 1.

1. Mark the signs and positions of both charges.
2. Predict attraction or repulsion before calculating magnitude.
3. For a ratio, compare the charge product and the square of the separation.
4. For a numerical force, convert charges and lengths to SI units first.

Recognising it. Use a ratio when a force is already known and only a charge or separation changes.

Pr 1. Two charges at distance r exert force F.

Solution. Keep both charges fixed and replace the separation r by 2r. The denominator becomes $(2r)^2 = 4r^2$, so the force becomes F/4. If instead one charge doubles at the original separation, its product with the other charge doubles, so force becomes 2F. Changing separation or charge magnitude alone does not change attraction into repulsion.

Common error. A negative charge does not mean a negative force magnitude.

In the examination. A magnitude answer is positive; state attraction or repulsion separately to communicate direction.

Pr 2. Tripling separation with fixed charges changes the force by what factor?

Pr 3. Two charges exert 12 N at separation r. What is the force at separation 2r, and at r if one charge is tripled?

Pr 4. Two unlike charges are moved farther apart without changing their signs. What changes?

(a) The attraction becomes weaker

(b) Attraction becomes repulsion

(c) The force keeps the same magnitude

2 Electric field and superposition

Aims. Combine electric fields by their directions at an observation point; Explain why a negative test charge feels force opposite the field.

Before this section: Vector addition; positive and negative charges; multiplication.

The electric field gives force per unit positive test charge. Field arrows point away from positive sources and toward negative sources. The field exists whether or not a test charge is present. Add fields as vectors, so equal magnitudes can cancel.

Idea 2. A field is a direction-and-strength instruction already present at a location. Its arrow says how a positive test charge would be pushed. An actual negative charge reads the same instruction with the direction reversed, so finding a field and finding a charge force are separate steps.

Fact 2.

$$E = F/q_{\text{test}}$$
$$E = k|Q|/r^2$$
$$\vec{F} = q\vec{E}$$

where E is the electric field magnitude (N/C); q is the charge experiencing the force (C); F is the force magnitude (N).

Why it holds. Multiplying by a negative charge reverses direction but does not make the force magnitude negative. The field is defined by the source arrangement and does not require the test charge to be present.

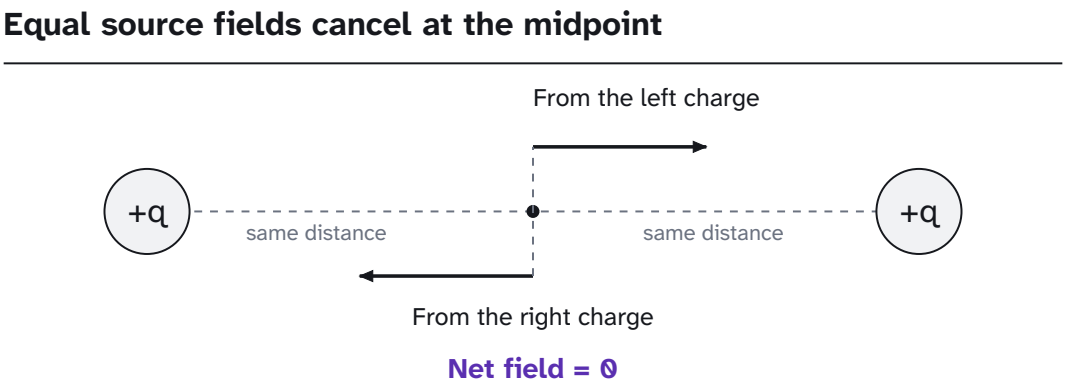


Figure 1. Figure 02. Equal positive charges give zero net field at the midpoint, so the force on a test charge there is zero. The two field arrows are offset vertically only for clarity; both represent fields at the midpoint.

Method 2.

1. Choose the observation point and draw the field arrow from each source there.
2. Point away from a positive source and toward a negative source.
3. Add signed collinear fields or use vector components when needed.
4. Only after finding the net field, use the test-charge sign to obtain force direction.

Recognising it. Separate field direction from force direction whenever the question supplies a negative charge.

Pr 5. At the midpoint between equal positive charges, the two electric fields oppose and cancel.

Solution. At the midpoint between equal positive charges, both source distances are the same, so field magnitudes match. The left positive charge points the field right; the right positive charge points it left. The net field is zero. Replace the right source by an equal negative charge: its field now points toward it, also to the right. The two fields therefore reinforce, pointing from the positive source toward the negative source.

Common error. Zero net field at a point does not mean each source field is zero.

In the examination. Draw field arrows before multiplying by a test charge; otherwise a negative test charge can lead to two direction reversals.

Pr 6. If the net field is zero at a point, is a charge placed there forced in either direction by that field?

Pr 7. A field of 200 N/C points east. Find the force on a charge of −0.003 C.

Pr 8. Two small test charges, one positive and one negative with equal magnitudes, are placed separately at the same point in a nonzero electric field. Their forces have:

(a) Different magnitudes solely because their signs differ

(b) Equal magnitudes and opposite directions

(c) Equal magnitudes and the same direction

3 Flux and Gauss law

Aims. Use the surface normal to determine electric flux; Distinguish enclosed charge from outside charge in Gauss law.
Before this section: Surface area; normal means perpendicular; cosine at 0° and 90°.

Electric flux measures the field passing through an oriented surface. The angle is measured from the surface normal, not the surface itself. Net flux through a closed surface depends on enclosed charge. An outside charge can affect the field at the surface but contributes zero net closed-surface flux.

Idea 3. Flux counts the field crossing a surface rather than the field lying along it. Imagine rain falling through a hoop: turning the hoop changes how much rain crosses its opening even when the rain itself stays the same. A surface normal records that orientation.

Fact 3.

$$\Phi = EA\cos\theta$$

$$\Phi_{\text{closed}} = Q_{\text{enclosed}}/\epsilon_0$$

where Φ is the electric flux ($\mathrm{N\,m^2/C}$); E is the electric field magnitude (N/C); A is the surface area ($\mathrm{m^2}$); Q_{enclosed} is the net charge inside a closed surface (C).

Why it holds. A field parallel to the normal crosses the full effective area. A field tangent to the surface has no crossing component, even if its magnitude is large.

Method 3.

1. Decide whether the question concerns one flat surface or a complete closed surface.
2. For a flat surface, draw the normal and identify its angle to the field.
3. For a closed surface, add only the enclosed charges with signs.
4. Interpret positive closed flux as net outward flux and negative as net inward flux.

Recognising it. Use flat-surface flux when area, a uniform field and orientation are given; use enclosed charge for net closed-surface flux.

Pr 9. A cube enclosing +q has net outward flux $\frac{q}{\varepsilon_0}$.

Solution. The cube is a closed surface enclosing net charge +q. Gauss law fixes its total outward flux as q divided by $\epsilon 0$. Moving q off-centre leaves enclosed charge unchanged, so total flux is unchanged. The nearby faces can now have stronger local fields; Gauss law does not say each face receives an equal share after the move.

Common error. Zero net closed-surface flux does not imply zero field everywhere.

In the examination. The angle in EA cos θ is to the normal; a field parallel to the surface gives zero flux.

Pr 10. Can an external charge produce a nonzero field on a box while contributing zero net flux through it?

Pr 11. A uniform field of 50 N/C passes through a flat area of 0.20 square metres. The chosen surface normal points in the same direction as the field. Find flux. What if the normal is perpendicular to the field?

Pr 12. A flat surface rotates from normal parallel to the field to field tangent to the surface. What happens to flux magnitude?

(a) It stays constant because field strength is unchanged

(b) It increases because the surface turns

(c) It falls from its maximum to zero

4 Symmetry and conductors

Aims. State the electrostatic properties of conducting material; Use the inverse-square field rule outside a spherical conductor.

Before this section: Electric field; conductor versus insulator; inverse-square ratios.

Gauss law is always valid, but it easily gives field magnitude only when symmetry makes the surface calculation simple. In electrostatic equilibrium the field inside conducting material is zero and excess charge lies on its surface. A conductor is an equipotential. For the outside-sphere inverse-square rule, assume an isolated conducting sphere with no external fields.

Idea 4. On an isolated conducting sphere, charge can rearrange until the conductor is in electrostatic equilibrium. Outside this symmetric arrangement, its field follows the same distance pattern as one central point charge. The relevant distance therefore starts at the centre, not at the surface.

Fact 4.

$$E_{\text{outside sphere}} = k|Q|/r^2$$
$$(r > R)$$

where Q is the net sphere charge (C); R is the sphere radius (m); r is the distance from sphere centre (m); E is the electric field magnitude (N/C).
Conditions. The sphere is isolated with spherical symmetry and is in electrostatic equilibrium.

Why it holds. Spherical symmetry makes all outside points at the same centre distance equivalent. The inverse-square outside field weakens by a factor of four when centre distance doubles.

Method 4.

1. Check that the conductor is in electrostatic equilibrium.
2. Distinguish a point in the conducting material from a point outside the sphere.
3. For an outside point measure distance from the centre, not from the surface.
4. Use the point-charge field or a simple ratio, then describe the radial direction.

Recognising it. Use the outside-sphere comparison only when the sphere is isolated as stated and both points lie outside it.

Pr 13. A charged conducting sphere behaves outside like a point charge at its centre.

Solution. Outside a spherically symmetric charged conducting sphere, the field has the same form as a charge at its centre. Compare two outside distances r and 2r. The square of the distance grows by a factor of four, so the field magnitude becomes E/4. This comparison uses centre distances; doubling the gap from the surface would generally give a different ratio.

Common error. Do not extend the zero-field-in-metal statement to every cavity containing a charge.

In the examination. Write r > R beside the outside-sphere formula to show that its domain has been checked.

Pr 14. A field is 80 N/C at 0.20 m from the centre. What is it at 0.40 m if both points are outside?

Pr 15. An isolated charged conducting sphere, with no external fields, has field magnitude 80 N/C at r = 0.20 m. Find the field at r = 0.40 m. Both distances are from the centre and outside the sphere.

Pr 16. Why can doubling the gap above a sphere surface differ from doubling the distance used in its field formula?

(a) The formula measures distance from the centre

(b) The formula always uses distance from the nearest charge on the surface

(c) Outside fields never depend on distance

5 Potential is not field

Aims. Add scalar potentials with their charge signs; Relate potential difference to energy change and field work.

Before this section: Scalar versus vector; signed energy changes; potential energy.

Electric potential is energy per unit charge, a scalar. Electric field is a vector. Add potentials with charge signs, but add fields with directions. Work by the electric field equals the decrease in electric potential energy.

Idea 5. Potential is a location-based energy-per-charge value, rather like height in a gravitational comparison. A change in potential tells how the potential energy of a particular charge changes. The electric field work has the opposite sign because energy given to kinetic motion comes from a decrease in potential energy.

Fact 5.

$$V = kQ/r \text{ (zero at infinity)}$$
$$\Delta U = q\Delta V$$
$$W_{\text{field}} = -\Delta U$$

where V is the electric potential (V = J/C); ΔV is the final minus initial potential (V); ΔU is the change in potential energy (J); q is the moving charge (C).

Why it holds. A positive charge moving to higher potential gains electric potential energy. Positive potential-energy gain means the electric field does negative work over that move.

Method 5.

1. Separate a field question, requiring direction, from a potential question, requiring signed scalar addition.
2. Choose the potential reference supplied by the problem.
3. Calculate energy change from charge times final-minus-initial potential.
4. Reverse the sign of that energy change to obtain work by the electric field.

Recognising it. When a question gives a charge and a potential difference, use the energy change directly rather than inventing a field or travel distance.

Pr 17. Equal positive charges can give zero field at their midpoint but positive potential there.

Solution. At the midpoint of two equal positive charges, equal field arrows point in opposite directions and cancel. Each positive charge contributes a positive potential at that point when zero potential is chosen at infinity. The two potentials add rather than cancel because potential has no directional arrow. Thus zero net field and positive potential can occur at the same point without contradiction.

Common error. A charge moving to lower potential does not always lose potential energy; a negative charge does the opposite.

In the examination. Write ΔV = Vfinal - Vinitial before calculating; swapping the endpoints reverses both energy and work signs.

Pr 18. A +2 C charge moves through ΔV = +3 V. What are ΔU and work by the field?

Pr 19. A +2 C charge moves through a potential difference of +3 V. Find its potential-energy change and the work done by the field.

Pr 20. A negative charge moves through a positive potential difference. What is the sign of its potential-energy change?

(a) Always zero

(b) Negative

(c) Positive

6 Capacitors and dielectrics

Aims. Distinguish series and parallel capacitor connections; Calculate stored charge and identify the effect of a dielectric.

Before this section: Potential difference; charge; reciprocals and addition.

A capacitor stores separated charge and electric-field energy. Capacitance describes charge stored per volt. Parallel capacitors share voltage and their capacitances add. Series capacitors carry equal charge magnitude and have a smaller equivalent capacitance.

Idea 6. Capacitance describes how much separated charge corresponds to each volt across two plates. A capacitor with a larger capacitance can store more charge at the same voltage. It is the connection to a battery or isolation of the capacitor that decides which quantity stays fixed during a change.

Fact 6.

$$Q = CV$$
$$U = \tfrac{1}{2}CV^2$$
$$C_{\parallel} = C_1 + C_2$$
$$1/C_s = 1/C_1 + 1/C_2$$

where C is the capacitance (F); Q is the stored charge magnitude on either plate (C); V is the potential difference between plates (V); U is the stored electric energy (J).

Why it holds. At fixed voltage, increasing capacitance increases stored charge. A microfarad multiplied by a volt gives a microcoulomb, so compatible prefixes can simplify a calculation.

Method 6.

1. Identify shared nodes: parallel capacitors share voltage, while ideal series capacitors carry equal charge magnitudes.
2. Add capacitances for parallel or add their reciprocals for series.
3. Use Q = CV with compatible prefixes such as μF times V giving μC.
4. For dielectric questions first check whether voltage or charge is held fixed.

Recognising it. Use the charge-capacitance-voltage relation when any two of those three quantities are supplied.

Pr 21. Two 6 μF capacitors give 12 μF in parallel and 3 μF in series.

Solution. For two 6 μF capacitors in parallel, capacitances add: $6 + 6 = 12$ μF. For series, $1/C_s = 1/6 + 1/6 = 1/3$ in reciprocal μF, so $C_s = 3$ μF. The series result is smaller than either individual capacitor, providing a quick plausibility check. Inserting a suitable dielectric increases capacitance; the resulting charge or voltage change depends on the connection to a battery.

Common error. Do not use resistor series/parallel rules for capacitors.

In the examination. For series, invert the reciprocal sum at the end; stopping at 1/C is a common lost result mark.

Pr 22. What charge magnitude is stored by 4 μF at 12 V?

Pr 23. A 4 μF capacitor is connected to 12 V. Find the charge in microcoulombs.

Pr 24. A capacitor stays connected to an ideal battery while its capacitance increases. What happens?

(a) Charge stays fixed and voltage increases

(b) Both voltage and charge must stay fixed

(c) Voltage stays fixed and stored charge increases

Fact 7. What changes a capacitor's capacitance?. These simple comparisons cover the geometry and dielectric ideas in the lecture without a derivation.

$$C = \frac{\varepsilon A}{d}$$

$$Q = CV$$

1. Use the first formula for parallel plates with a small gap compared with plate size, so edge effects can be ignored.
2. Here A is overlapping plate area, d is separation, and ε describes the material filling the gap.
3. Doubling area doubles capacitance. Doubling separation halves it.
4. A dielectric filling the gap increases capacitance. Battery attached: voltage stays fixed. Disconnected and isolated: charge stays fixed.

Pr 25. An isolated charged capacitor has capacitance 2 μF and voltage 12 V. A dielectric triples its capacitance. Find the new charge and voltage.

7 Current, resistance and electrical power

Aims. Find current, voltage and power for an ohmic resistor; Predict how wire length and area change resistance.

Before this section: Charge flow; multiplication and division; circuit voltage.

Current is the rate of charge flow. Conventional current follows positive-charge flow, opposite electron drift in a metal. Resistance depends on material and geometry. Ohm law describes an ohmic element under fixed physical conditions.

Idea 7. Voltage is the energy transferred per unit charge, while current counts charge passing each second. Multiplying them gives energy per second, which is power. Resistance connects the current through a resistor with the voltage across that same resistor; those locations matter in a circuit.

Fact 8.

$$I = \Delta Q/\Delta t$$
$$V = IR$$
$$R = \rho L/A$$
$$P = VI$$

where I is the conventional current (A); R is the resistance (Ω); V is the potential difference (V); P is the electrical power (W); ρ is the resistivity (Ω m).
Conditions. Ohm law is used for an ohmic resistor under unchanged physical conditions.

Why it holds. For a given resistor current, the voltage drop follows its resistance. Power follows both the amount of charge passing each second and the energy transferred by each unit charge.

Method 7.

1. Label voltage across the resistor and current through it.
2. Use V = IR to find the missing electrical quantity.
3. Calculate power using VI or an equivalent resistor relation.
4. For wire comparisons hold material and temperature fixed before changing length or area.

Recognising it. When current and resistance are supplied, find the resistor voltage first, then use voltage times current for power.

Pr 26. A 6 Ω resistor across 12 V carries 2 A and dissipates 24 W.

Solution. The 6 Ω resistor has 12 V across it, so I = 12/6 = 2 A. Its electrical power is VI = 12 × 2 = 24 W. Power of 24 W means 24 joules of energy are transferred each second. A longer wire with unchanged material, area and temperature has larger resistance because R is proportional to length.

Common error. More resistance lowers power at fixed voltage but raises power at fixed current.

In the examination. Do not confuse power in watts with energy in joules; multiply power by time when energy is requested.

Pr 27. What happens to resistance if a wire length doubles but material, area and temperature stay fixed?

Pr 28. A 3 Ω resistor carries 2 A. Find voltage and power.

Pr 29. The current through the same fixed resistor doubles. What happens to its power?

(a) It becomes four times as large

(b) It doubles

(c) It stays fixed because resistance stays fixed

8 DC networks, Kirchhoff rules and RC behaviour

Aims. Reduce simple resistor networks and recover branch currents; Describe the initial and final behaviour of an RC charging circuit.

Before this section: Ohm law; series and parallel connections; charge and energy conservation.

Series resistors share current; parallel resistors share voltage. Kirchhoff junction rules express charge conservation, and loop rules express energy conservation. An initially uncharged capacitor charges gradually through a resistor and eventually blocks steady DC in the ideal model.

Idea 8. In a series path, charge has no branch in which to split, so the same current passes through each resistor. The source voltage is shared as energy drops along the path. A larger resistance takes a larger voltage share because the current is common.

Fact 9.

$$R_s = R_1 + R_2$$
$$1/R_p = 1/R_1 + 1/R_2$$
$$\tau = RC$$

where R_{eq} is the equivalent resistance (Ω); I is the current (A); V is the voltage (V); τ is the RC time constant (s).

Why it holds. Series resistances add because their voltage drops add at the same current. Checking that the individual drops sum to the source voltage tests the energy bookkeeping.

Method 8.

1. Identify series elements by shared current and parallel elements by shared voltage.
2. Reduce the network and calculate total source current.
3. Return to the original circuit to calculate branch currents or voltage drops.
4. Check junction current sums and loop voltage sums; for RC questions distinguish transient from long-time behaviour.

Recognising it. Use series reduction when every charge passing one element must pass the next without a junction.

Pr 30. Two 6 Ω resistors in parallel give 3 Ω.

Solution. Both 6 Ω resistors are connected across the same two nodes, so they are parallel. Their equal-resistor equivalent is $6/2 = 3\ \Omega$. The 12 V source therefore supplies $12/3 = 4$ A total. Each branch still has 12 V and carries $12/6 = 2$ A; the branch currents sum to the 4 A source current.

Common error. Adjacent components in a drawing are not necessarily in series.

In the examination. After finding an equivalent resistance, redraw or relabel the original circuit before assigning individual voltage drops.

Pr 31. After a long time, does an ideal capacitor in a DC series RC circuit carry a steady current?

Pr 32. A 4 Ω and 8 Ω resistor are in series across 24 V. Find total current and each voltage drop.

Pr 33. A 4-ohm resistor and an 8-ohm resistor carry the same series current. Which has the larger voltage drop?

(a) The drops must be equal because current is equal

(b) The 8-ohm resistor

(c) The 4-ohm resistor

Fact 10. One time constant: the 63% and 37% rules. Recognize how quickly a capacitor charges or discharges without deriving an exponential.

$$\tau = RC$$

$$1 - e^{-1} \approx 0.632$$

$$e^{-1} \approx 0.368$$

1. Use an ideal series resistor-capacitor circuit with constant R and C. The time constant τ is measured in seconds.
2. Charging from zero: after one time constant, capacitor voltage and charge reach about 63% of their final values.
3. Discharging: after one time constant, about 37% of the starting voltage and charge remain.
4. A larger resistance or capacitance makes the change slower.

Pr 34. A capacitor charges from zero toward 10 V. Its time constant is 2 s. Estimate its voltage after 2 s.

Fact 11. Kirchhoff signs and a real battery. Choosing consistent signs is the main skill. A negative current answer tells you the actual direction.

$$\text{current in} = \text{current out}$$

$$\sum \Delta V = 0$$

$$V_{\text{terminal}} = \mathcal{E} - Ir \quad \text{(battery delivering current)}$$

1. At a junction, charge does not keep building up: add entering and leaving currents separately.
2. Choose a loop direction. Across a resistor, travelling with the assumed current gives $-IR$; travelling against it gives $+IR$.
3. Across a battery, going from its negative to positive terminal gives $+\mathcal{E}$; the reverse gives $-\mathcal{E}$.
4. For a battery supplying current, internal resistance r causes the terminal voltage to be less than its emf $\mathcal{E}$.

Pr 35. A battery has emf 12 V and internal resistance 1 Ω. It supplies a 5 Ω resistor. Find current and terminal voltage.

9 Magnetic force: direction before calculation

Aims. Determine when magnetic force vanishes or is largest; Explain circular motion without a change in kinetic energy.
Before this section: Perpendicular directions; circular acceleration; charge sign.

A magnetic field exerts force on a moving charge when its motion has a perpendicular component. The force is perpendicular to velocity and therefore does no work on a point charge. It can change direction without changing speed.

Idea 9. Magnetic force acts sideways relative to moving charge. For a straight current-carrying wire, the current direction plays the corresponding role. Geometry matters: a wire across the field has maximum magnetic force, whereas a wire along the field has no perpendicular component and no magnetic force.

Fact 12.

$$F = |q|vB\sin\theta$$
$$F_{\text{wire}} = BIL\sin\theta$$

where q is the particle charge (C); v is the particle speed (m/s); B is the magnetic field magnitude (T); F is the magnetic force magnitude (N); θ is the angle between velocity and field (degrees or radians).

Why it holds. A perpendicular arrangement uses the full field-current-length product because the sine factor is one. Reversing current reverses force direction but does not change magnitude when other factors stay fixed.

Method 9.

1. Decide whether the object is a moving charge or a current-carrying wire.
2. Identify the angle to the magnetic field and the perpendicular motion or wire length.
3. Use the right-hand rule for positive charge or conventional current; reverse for negative charge.
4. Calculate magnitude and check that the force is perpendicular to motion.

Recognising it. Use the wire-force relation when current, straight wire length and magnetic field are given; do not substitute particle speed.

Pr 36. A positive charge moving perpendicular to a uniform magnetic field follows a circular path if no other force acts.

Solution. For the positive charge in the example, velocity is perpendicular to the uniform field, so the magnetic force magnitude is qvB. At each point the force is perpendicular to velocity and turns the particle inward. Because the force does no work, speed and kinetic energy remain constant. With no other forces, the continual turning gives a circular path in the plane perpendicular to the field.

Common error. Do not apply the right-hand direction unchanged to an electron.

In the examination. A direction answer needs the charge sign; a right-hand-rule result for a positive charge must be reversed for an electron.

Pr 37. Does a charge moving exactly along a magnetic field feel magnetic force?

Pr 38. A straight 0.50 m wire carries 2 A perpendicular to a 0.30 T field. Find force magnitude.

Pr 39. A straight wire initially lies perpendicular to a uniform field. It rotates until parallel while current stays fixed. Its magnetic force magnitude:

(a) Stays unchanged

(b) Doubles

(c) Falls to zero

10 Sources of magnetic fields

Aims. Relate a current to the direction of the magnetic field it produces; Use simple distance and current ratios for a long wire.
Before this section: Current; inverse proportionality; circular field-line diagrams.

Currents produce magnetic fields. Around a long straight wire the field forms circles. Inside an ideal long solenoid the field is approximately uniform. For two parallel wires, currents in the same direction attract and opposite currents repel.

Idea 10. A straight current-carrying wire has magnetic field lines that circle the wire. Moving farther away weakens the field, but its distance pattern differs from the field of a point charge. Recognizing the source geometry prevents using the familiar but incorrect inverse-square rule here.

Fact 13.

$$B_{\text{wire}} = \mu_0 I/(2\pi r)$$
$$B_{\text{solenoid}} = \mu_0 n I$$

where B is the magnetic field magnitude (T); I is the wire current (A); r is the perpendicular distance from wire (m); n is the solenoid turns per unit length (m^{-1}).
Conditions. The wire is sufficiently long and the observation point sufficiently far from its ends for the long-wire approximation.

Why it holds. For an ideal long wire, magnetic field magnitude is inversely proportional to perpendicular distance. At unchanged current, a threefold increase in distance therefore gives a threefold decrease rather than a ninefold decrease.

Method 10.

1. Identify whether the source is a straight wire, a solenoid or a pair of wires.

2. For a straight wire, use the right-hand grip rule to draw field circles.
3. Apply the matching field relation; for ratios hold unstated factors fixed.
4. For paired wires determine attraction or repulsion from relative current directions.

Recognising it. The phrase long straight wire signals the current-over-distance rule, not the point-charge inverse-square rule.

Pr 40. Doubling distance from a long wire halves its magnetic field; doubling current doubles the field.

Solution. For the long wire, B is proportional to I/r. Moving from r to 2r at the same current gives half the original field. Doubling current at fixed r instead doubles the field. If current and distance both double, the two factors cancel and field magnitude stays the same.

Common error. The wire field varies as inverse distance, not inverse distance squared.

In the examination. Use $1/r$ for the long-wire field; the $1/r^2$ rule belongs to a different source geometry.

Pr 41. A wire gives 30 μT at 2 cm. What field does it give at 6 cm with the same current?

Pr 42. The field is 30 microtesla at 2 cm from a long straight wire. Find the field at 6 cm at the same current.

Pr 43. Current and perpendicular distance from a long wire both double. What happens to magnetic field magnitude?

(a) It stays unchanged

(b) It doubles

(c) It halves

11 Induction and Lenz law

Aims. Identify which change produces an induced emf; Use Lenz law to choose the induced current direction.

Before this section: Magnetic field direction; loop area; clockwise versus counterclockwise viewed from a stated side.

An induced emf requires a change in magnetic flux, which can arise from changing field, area or orientation. Lenz law sets the direction: the induced effect opposes the change of flux. A static magnet near a stationary loop does not produce a sustained induced current.

Idea 11. Induction responds to a changing magnetic flux through a loop, rather than merely to the presence of a magnet. A faster change gives a larger average emf. A coil repeats the same flux-change effect through many turns, so their induced emfs add when the turns are linked alike.

Fact 14.

$$|\mathcal{E}| = N|\Delta\Phi_B|/\Delta t$$
$$\Phi_B = BA\cos\theta$$

where Φ_B is the magnetic flux through one turn (Wb); N is the number of coil turns (dimensionless); $\mathcal{E}$ is the induced emf (V); Δt is the time interval of flux change (s).

Why it holds. More turns multiply the emf when each turn experiences the same flux change. Spreading the same flux change over a longer time reduces the average emf.

Method 11.

1. State the original flux direction and whether its magnitude increases or decreases.
2. Choose an induced field that opposes that change.
3. Use the grip rule to convert the required field to current direction, naming the viewing side.
4. For magnitude, multiply the flux change per turn by the number of turns and divide by elapsed time.

Recognising it. Look for a change in field, loop area or loop orientation over a stated time; identify whether flux is per turn.

Pr 44. If outward flux increases, the induced field points inward.

Solution. In the example, outward flux through the loop is increasing. The induced effect must oppose the increase, so its own field points inward. Viewed from the side toward which the original flux points outward, the inward field requires clockwise current. If the outward flux were decreasing instead, the induced field and current direction would both reverse.

Common error. Opposes the change is not the same as always opposes the original field.

In the examination. Say oppose the change in flux, not oppose the flux: a decreasing field produces a reinforcing induced field.

Pr 45. Does a stationary loop near an unchanging stationary magnet have a sustained induced emf?

Pr 46. A 20-turn coil has flux per turn changing by 0.03 Wb in 0.10 s. Find average emf magnitude.

Pr 47. The same flux change per turn occurs in half the time. With turn count unchanged, average emf magnitude:

(a) Stays unchanged

(b) Doubles

(c) Halves

12 Inductance, AC and electromagnetic waves

Aims. Convert sinusoidal peak voltage to RMS and calculate resistive heating; Relate electromagnetic frequency to vacuum wavelength.
Before this section: Square roots; frequency and wavelength; electrical power.

Course-plan extension: an inductor resists changes of current and stores magnetic energy. AC reverses direction periodically. RMS values describe equivalent heating for resistive loads. In vacuum an EM wave has mutually perpendicular electric field, magnetic field and travel direction.

Idea 12. An alternating voltage has a changing instantaneous value, so its peak is not its steady heating equivalent. RMS voltage compresses that varying waveform into the DC voltage that would produce the same average heating in a resistor. The familiar peak conversion depends on the waveform being sinusoidal.

Fact 15.

$$U_L = \tfrac{1}{2}LI^2$$
$$V_{\text{rms}} = V_{\text{peak}}/\sqrt{2}\ \text{(sinusoidal)}$$
$$c = f\lambda\ \text{(vacuum)}$$

where V_{peak} is the maximum sinusoidal voltage magnitude (V); V_{rms} is the root-mean-square voltage (V); L is the inductance (H); f is the wave frequency (Hz); λ is the wavelength (m). *Conditions.* The peak-to-RMS conversion assumes a sinusoid; the wavelength calculation assumes propagation in vacuum.

Why it holds. Resistive heating depends on voltage squared, so simply averaging positive and negative voltage would miss the heating. A sinusoid has RMS voltage below its peak because it spends only part of each cycle near the peak.

Method 12.

1. Identify whether the question concerns an inductor, sinusoidal AC or an electromagnetic wave.
2. For sinusoidal AC divide peak voltage by the square root of two.
3. Use RMS values for average power in a pure resistor.
4. For vacuum waves calculate wavelength as c divided by frequency and convert metres to the requested unit.

Recognising it. Use RMS values for average resistive power; use the peak-to-RMS conversion only for a specified sinusoid.

Pr 48. A sinusoidal supply with peak voltage 141.4 V has RMS voltage 100 V.

Solution. For the stated sinusoidal supply, 141.4 divided by $\sqrt{2}$ is approximately 100 V RMS. This RMS value matches the heating effect of 100 V DC in the same pure resistor.

For the separate light example, wavelength is 3.00×10^8 divided by $6 \times 10^{14} = 5 \times 10^{-7}$ m. Since one nanometre is 10^{-9} m, the wavelength is 500 nm.

Common error. An ideal transformer does not continuously transform steady DC after switching transients.

In the examination. Keep this extension conceptual: use RMS for resistor power and $c = f\lambda$ for waves; do not mix peak and RMS values.

Pr 49. A 20 V peak sinusoid is applied to 10 Ω. What is the average power?

Pr 50. A sinusoidal voltage has peak value 20 V. Find its RMS value, then the average power in a 10 Ω resistor.

Pr 51. A symmetric sinusoidal voltage has zero average voltage over a cycle. Does a resistor receive zero average power?

(a) Yes; positive and negative heating cancel

(b) Only the positive half-cycle heats it

(c) No; both voltage polarities produce heating

Hints

Use one hint at a time, then return to the problem.

Pr 3 (p. 3) (1) Treat the two requested situations separately, each starting from the original 12 N force. (2) For doubled separation, square the separation-change factor before applying it to the force. (3) For one charge tripled at the original separation, apply only the charge-change factor. State both results with newtons.

Pr 7 (p. 5) (1) The field direction is east. Decide the force direction using the sign of the supplied charge before calculating. (2) Multiply the field magnitude 200 N/C by the charge magnitude 0.003 C. (3) State the positive force magnitude in newtons and its compass direction. The minus sign belongs to the directional reasoning, not a magnitude.

Pr 11 (p. 6) (1) Draw the chosen normal as an arrow. In the first case it points with the field, so identify their angle. (2) Multiply field 50 N/C by area 0.20 m^2 and the cosine of that normal angle. (3) For the second case change only the cosine factor. Keep flux units distinct from field units.

Pr 15 (p. 7) (1) Check that both stated distances start at the centre and exceed the sphere radius as specified. (2) Compare 0.40 m with 0.20 m and square that distance ratio. (3) Divide the original 80 N/C by the squared distance factor. No unknown total charge is needed.

Pr 19 (p. 9) (1) Treat the stated +3 V as final potential minus initial potential. Identify the moving charge sign. (2) Multiply charge +2 C by the potential difference to obtain the signed potential-energy change. (3) Reverse that energy-change sign for work done by the field. Report each quantity in joules and label them separately.

Pr 23 (p. 10) (1) The capacitor is connected across 12 V. Identify whether charge follows by multiplying or dividing capacitance and voltage. (2) Use capacitance 4 microfarads and the supplied voltage. (3) A result from microfarads times volts is in microcoulombs. If using farads instead, convert the final coulombs back consistently.

Pr 25 (p. 11) (1) Initial charge: $Q = CV = 2 \times 12 = 24\,\mu$C. (2) Isolation keeps the charge at 24 μC. (3) New capacitance: 6 μF; new voltage: $V = 24/6 = 4$ V.

Pr 28 (p. 12) (1) Mark the given 2 A as current through the 3-ohm resistor. Which relation gives its voltage drop? (2) Multiply current by resistance to obtain voltage. (3) Use that voltage with the same current to obtain power. Check that the units are volts for the first result and watts for the second.

Pr 32 (p. 13) (1) Confirm there is one series path. Add the two resistances before finding source current. (2) Divide the 24 V source voltage by the total resistance. (3) Use that same current with each resistance separately. Verify that the two voltage drops add to 24 V.

Pr 34 (p. 14) (1) The elapsed time is one time constant. (2) Multiply final voltage by 0.632: $10 \times 0.632 = 6.32$ V.

Pr 35 (p. 14) (1) Use one loop: $12 - I(1) - I(5) = 0$. (2) Total resistance is 6 Ω, so $I = 12/6 = 2$ A. (3) Terminal voltage is $12 - (2)(1) = 10$ V. Check the external resistor: $IR = (2)(5) = 10$ V.

Pr 38 (p. 16) (1) This is a current-carrying wire, not a single moving particle. Choose the relation involving field, current and length. (2) The wire is perpendicular to the field, so

the angle factor is one. (3) Multiply 0.30 T, 2 A and 0.50 m. The question asks for magnitude, so no charge sign is needed.

Pr 42 (p. 17) (1) Keep the current fixed as stated. Compare the new perpendicular distance with the original distance. (2) Use an inverse-distance ratio for a long wire; do not square the ratio. (3) Apply that factor to 30 microtesla. Centimetres can remain in the ratio because both distances use the same unit.

Pr 46 (p. 18) (1) The problem gives flux change per turn. Decide where the 20 turns enter the total effect. (2) Multiply turn count by flux change per turn, then divide by the 0.10 s interval. (3) Use the magnitude of the change because magnitude is requested. A current direction would require separate Lenz-law information.

Pr 50 (p. 20) (1) Separate the request into RMS voltage and average resistor power. The supplied 20 V is a peak value. (2) Divide the peak by the square root of two to obtain the sinusoidal RMS value. (3) Square that RMS voltage and divide by 10 ohms. Keep a few extra digits until the final rounding.

Answers

Pr 2 (p. 3) It becomes one ninth as large.

Pr 3 (p. 3) At 2r: 12/4 = 3 N. Tripling one charge at r: 36 N. Two marks for each ratio and result.

Pr 4 (p. 3) (a) Correct. Opposite signs still attract, but larger separation reduces magnitude.

Pr 6 (p. 4) No. The electric force qE is zero there in the stated source-field model.

Pr 7 (p. 5) Magnitude = 0.003 × 200 = 0.60 N. Direction is west because the charge is negative. Two marks for magnitude and two for direction with reasoning.

Pr 8 (p. 5) (b) Correct. The field is the same and charge magnitudes match; opposite signs reverse force direction.

Pr 10 (p. 6) Yes. Field enters and leaves; the total closed-surface flux from that external charge cancels.

Pr 11 (p. 6) Parallel normal: flux = 50 × 0.20 = 10 N m squared/C. Perpendicular normal: cosine 90 degrees = 0, so flux = 0. Two marks per case.

Pr 12 (p. 6) (c) Correct. The crossing component decreases to zero as the normal becomes perpendicular to the field.

Pr 14 (p. 7) 20 N/C: the centre distance doubles and the field is divided by four.

Pr 15 (p. 7) Inverse-square scaling gives 80 × (0.20/0.40) squared = 20 N/C. Credit the outside-sphere condition and inverse-square reasoning.

Pr 16 (p. 7) (a) Correct. Centre distance equals sphere radius plus the surface gap, so the two doubling operations differ.

Pr 18 (p. 9) $\Delta U = +6$ J and work by the field $= -6$ J.

Pr 19 (p. 9) Energy change = 2 × 3 = +6 J. Work by field = −6 J. Two marks each; the signs are essential.

Pr 20 (p. 9) (b) Correct. Charge times potential difference is negative for a negative charge and positive difference.

Pr 22 (p. 10) 48 μC, from Q = CV.

Pr 23 (p. 10) Q = CV = 4 × 12 = 48 microcoulombs. In SI this is 0.000048 C. Credit formula, substitution, prefix and result.

Pr 24 (p. 10) (c) Correct. The battery fixes voltage and supplies the extra charge associated with larger capacitance.

Pr 25 (p. 11) Charge stays 24 μC; voltage falls to 4 V.

Pr 27 (p. 12) Resistance doubles.

Pr 28 (p. 12) V = IR = 6 V. P = VI = 12 W (or I^2R). Two marks per result.

Pr 29 (p. 12) (a) Correct. Both current and its corresponding voltage double, so their product quadruples.

Pr 31 (p. 13) No. Once charged, it blocks steady DC in the ideal model.

Pr 32 (p. 13) Total resistance = 12 Ω; current = 2 A; drops are 8 V and 16 V. One mark each; the drops must sum to 24 V.

Pr 33 (p. 13) (b) Correct. At common current, voltage drop is proportional to resistance.

Pr 34 (p. 14) About 6.3 V. It is not fully charged after one time constant.

Pr 35 (p. 14) Current 2 A; terminal voltage 10 V.

Pr 37 (p. 16) No. The angle is 0° or 180°, so its sine is zero.

Pr 38 (p. 16) F = BIL = 0.30 × 2 × 0.50 = 0.30 N. Four marks for formula, perpendicular condition, substitution and unit.

Pr 39 (p. 16) (c) Correct. At parallel alignment the perpendicular component and sine factor vanish.

Pr 41 (p. 17) 10 μT because distance triples.

Pr 42 (p. 17) B is inversely proportional to r, so B = 30 × 2/6 = 10 microtesla. Credit model, ratio, arithmetic and unit.

Pr 43 (p. 17) (a) Correct. Doubling current and distance multiplies the numerator and denominator by the same factor.

Pr 45 (p. 18) No. Its magnetic flux is not changing.

Pr 46 (p. 18) Magnitude = 20 × 0.03/0.10 = 6 V. The direction needs the direction of the flux change; it cannot be inferred from these magnitudes alone.

Pr 47 (p. 18) (b) Correct. Emf depends on the rate of flux change, so halving time doubles that rate.

Pr 49 (p. 20) 20 W: $V_{rms}^2/R = (20^2/2)/10$.

Pr 50 (p. 20) RMS = 20/square root of 2 = 14.14 V. Average power = RMS squared/10 = 20 W. Two marks per stage.

Pr 51 (p. 20) (c) Correct. The heating depends on voltage squared, which is nonnegative for either polarity.

Electricity and Magnetism, Test A

30 marks. Suggested time 30 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. A neutral object loses electrons. It becomes:

   (a) Negative
   (b) Positive
   (c) Uncharged forever
   (d) Magnetic only

2. A negative test charge in a field directed right feels force:

   (a) Left
   (b) Zero
   (c) Right
   (d) Up

3. A charge outside a closed empty box gives net flux through the box of:

   (a) $\frac{q}{\varepsilon_0}$
   (b) Always positive
   (c) Always negative
   (d) Zero

4. Inside the material of a conductor in electrostatic equilibrium, the electric field is:

   (a) Equal to its surface value
   (b) Infinite
   (c) Zero
   (d) Always outward

5. For equal and opposite point charges, take potential as zero at infinity. At their midpoint the potential is:

   (a) Zero
   (b) Undefined
   (c) Positive
   (d) Negative

6. Adding a dielectric to an isolated charged capacitor increases C. Its voltage:

   (a) Decreases
   (b) Increases
   (c) Must stay fixed
   (d) Becomes infinite

Part II. Show your reasoning.

1. Two charges exert 12 N at separation r. What is the force at separation 2r, and at r if one charge is tripled?

2. A field of 200 N/C points east. Find the force on a charge of -0.003 C.

3. A uniform field of 50 N/C passes through a flat area of 0.20 square metres. The chosen surface normal points in the same direction as the field. Find flux. What if the normal is perpendicular to the field?

4. An isolated charged conducting sphere, with no external fields, has field magnitude 80 N/C at $r = 0.20$ m. Find the field at $r = 0.40$ m. Both distances are from the centre and outside the sphere.

5. A +2 C charge moves through a potential difference of +3 V. Find its potential-energy change and the work done by the field.

6. A 4 μF capacitor is connected to 12 V. Find the charge in microcoulombs.

Electricity and Magnetism, Test B

30 marks. Suggested time 30 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. At fixed voltage, doubling resistance makes power:

   (a) Unchanged
   (b) Double
   (c) Half
   (d) Four times

2. After a long time in an ideal DC charging circuit, capacitor current is:

   (a) Maximum
   (b) Negative forever
   (c) Zero
   (d) Equal to the battery voltage divided by the resistor resistance

3. A charge moving parallel to a magnetic field experiences magnetic force:

   (a) qvB
   (b) Twice qvB
   (c) Opposite its velocity
   (d) Zero

4. Two parallel wires carrying currents in the same direction:

   (a) Repel
   (b) Attract
   (c) Have no interaction
   (d) Must carry zero field

5. Which situation produces an induced emf?

   (a) No change of area, field or angle
   (b) Changing magnetic flux
   (c) A stationary unchanging field only
   (d) Constant flux

6. In vacuum, electric and magnetic fields of a plane EM wave are:

   (a) Parallel to each other
   (b) Longitudinal only
   (c) Zero
   (d) Perpendicular to each other and to travel

Part II. Show your reasoning.

1. A 3 Ω resistor carries 2 A. Find voltage and power.

2. A 4 Ω and 8 Ω resistor are in series across 24 V. Find total current and each voltage drop.

3. A straight 0.50 m wire carries 2 A perpendicular to a 0.30 T field. Find force magnitude.

4. The field is 30 microtesla at 2 cm from a long straight wire. Find the field at 6 cm at the same current.

5. A 20-turn coil has flux per turn changing by 0.03 Wb in 0.10 s. Find average emf magnitude.

6. A sinusoidal voltage has peak value 20 V. Find its RMS value, then the average power in a 10 Ω resistor.

Electricity and Magnetism, Test A: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (b) Removing negative charge leaves a positive net charge.

Written part. At 2r: 12/4 = 3 N. Tripling one charge at r: 36 N. Two marks for each ratio and result. (Section: p. 2.)

Question 2. (a) The negative charge reverses the field direction in $F = qE$.

Written part. Magnitude = 0.003 × 200 = 0.60 N. Direction is west because the charge is negative. Two marks for magnitude and two for direction with reasoning. (Section: p. 3.)

Question 3. (d) No charge is enclosed, so total entering and leaving flux cancels.

Written part. Parallel normal: flux = 50 × 0.20 = 10 N m squared/C. Perpendicular normal: cosine 90 degrees = 0, so flux = 0. Two marks per case. (Section: p. 5.)

Question 4. (c) Free charges redistribute until the internal field is cancelled.

Written part. Inverse-square scaling gives 80 × (0.20/0.40) squared = 20 N/C. Credit the outside-sphere condition and inverse-square reasoning. (Section: p. 6.)

Question 5. (a) The signed scalar potentials cancel when distances and charge magnitudes match.

Written part. Energy change = 2 × 3 = +6 J. Work by field = −6 J. Two marks each; the signs are essential. (Section: p. 8.)

Question 6. (a) Charge remains fixed and $V = \frac{Q}{C}$.

Written part. Q = CV = 4 × 12 = 48 microcoulombs. In SI this is 0.000048 C. Credit formula, substitution, prefix and result. (Section: p. 9.)

Electricity and Magnetism, Test B: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (c) $P = \frac{V^2}{R}$, so doubling the denominator halves it.

Written part. V = IR = 6 V. P = VI = 12 W (or I^2R). Two marks per result. (Section: p. 11.)

Question 2. (c) After a long time, the capacitor voltage equals the battery voltage. The resistor voltage and circuit current are zero. During charging, charge moves in the wires onto and off the plates; it does not cross the ideal dielectric.

Written part. Total resistance = 12 Ω; current = 2 A; drops are 8 V and 16 V. One mark each; the drops must sum to 24 V. (Section: p. 12.)

Question 3. (d) The sine of the angle is zero.

Written part. F = BIL = 0.30 × 2 × 0.50 = 0.30 N. Four marks for formula, perpendicular condition, substitution and unit. (Section: p. 15.)

Question 4. (b) Each current experiences the magnetic field of the other wire.

Written part. B is inversely proportional to r, so B = 30 × 2/6 = 10 microtesla. Credit model, ratio, arithmetic and unit. (Section: p. 16.)

Question 5. (b) Faraday law depends on the rate of flux change.

Written part. Magnitude = 20 × 0.03/0.10 = 6 V. The direction needs the direction of the flux change; it cannot be inferred from these magnitudes alone. (Section: p. 17.)

Question 6. (d) A plane electromagnetic wave in vacuum is transverse.

Written part. RMS = 20/square root of 2 = 14.14 V. Average power = RMS squared/10 = 20 W. Two marks per stage. (Section: p. 19.)

Facts at a glance

1. Charges, conductors and Coulomb force p. 2

$$F = k|q_1 q_2|/r^2$$
$$k = 8.99 \times 10^9 \ \mathrm{N\,m^2/C^2}$$

2. Electric field and superposition p. 3

$$E = F/q_{\text{test}}$$
$$E = k|Q|/r^2$$
$$\vec{F} = q\vec{E}$$

3. Flux and Gauss law p. 5

$$\Phi = EA\cos\theta$$
$$\Phi_{\text{closed}} = Q_{\text{enclosed}}/\epsilon_0$$

4. Symmetry and conductors p. 6

$$E_{\text{outside sphere}} = k|Q|/r^2$$
$$(r > R)$$

5. Potential is not field p. 8

$$V = kQ/r \ \text{(zero at infinity)}$$
$$\Delta U = q\Delta V$$
$$W_{\text{field}} = -\Delta U$$

6. Capacitors and dielectrics p. 9

$$Q = CV$$
$$U = \tfrac{1}{2}CV^2$$
$$C_{\parallel} = C_1 + C_2$$
$$1/C_s = 1/C_1 + 1/C_2$$

7. Current, resistance and electrical power p. 11

$$I = \Delta Q/\Delta t$$
$$V = IR$$
$$R = \rho L/A$$
$$P = VI$$

8. DC networks, Kirchhoff rules and RC behaviour p. 12

$$R_s = R_1 + R_2$$
$$1/R_p = 1/R_1 + 1/R_2$$
$$\tau = RC$$

9. Magnetic force: direction before calculation p. 15

$$F = |q|vB\sin\theta$$
$$F_{\text{wire}} = BIL\sin\theta$$

10. Sources of magnetic fields p. 16

$$B_{\text{wire}} = \mu_0 I/(2\pi r)$$
$$B_{\text{solenoid}} = \mu_0 nI$$

11. Induction and Lenz law p. 17

$$|\mathcal{E}| = N|\Delta\Phi_B|/\Delta t$$
$$\Phi_B = BA\cos\theta$$

12. Inductance, AC and electromagnetic waves p. 19

$$U_L = \tfrac{1}{2}LI^2$$
$$V_{\text{rms}} = V_{\text{peak}}/\sqrt{2}\ \text{(sinusoidal)}$$
$$c = f\lambda\ \text{(vacuum)}$$

Notation

Symbols are listed by section. The same letter may name different quantities in different sections; read the section's Fact.

1. Charges, conductors and Coulomb force

F	electrostatic force magnitude (N)
q_1, q_2	source charges (C)
r	separation of charges (m)

2. Electric field and superposition

E	electric field magnitude (N/C)
q	charge experiencing the force (C)
F	force magnitude (N)

3. Flux and Gauss law

Φ	electric flux (N m^2/C)
E	electric field magnitude (N/C)
A	surface area (m^2)
Q_{enclosed}	net charge inside a closed surface (C)

4. Symmetry and conductors

Q	net sphere charge (C)
R	sphere radius (m)
r	distance from sphere centre (m)
E	electric field magnitude (N/C)

5. Potential is not field

V	electric potential (V = J/C)
ΔV	final minus initial potential (V)
ΔU	change in potential energy (J)
q	moving charge (C)

6. Capacitors and dielectrics

C	capacitance (F)
Q	stored charge magnitude on either plate (C)
V	potential difference between plates (V)
U	stored electric energy (J)

7. Current, resistance and electrical power

I	conventional current (A)
R	resistance (Ω)
V	potential difference (V)
P	electrical power (W)
ρ	resistivity (Ω m)

8. DC networks, Kirchhoff rules and RC behaviour

R_{eq}	equivalent resistance (Ω)
I	current (A)
V	voltage (V)
τ	RC time constant (s)

9. Magnetic force: direction before calculation

q	particle charge (C)
v	particle speed (m/s)
B	magnetic field magnitude (T)
F	magnetic force magnitude (N)
θ	angle between velocity and field (degrees or radians)

10. Sources of magnetic fields

B	magnetic field magnitude (T)
I	wire current (A)
r	perpendicular distance from wire (m)
n	solenoid turns per unit length (m^{-1})

11. Induction and Lenz law

Φ_B	magnetic flux through one turn (Wb)
N	number of coil turns (dimensionless)
$\mathcal{E}$	induced emf (V)
Δt	time interval of flux change (s)

12. Inductance, AC and electromagnetic waves

V_{peak}	maximum sinusoidal voltage magnitude (V)
V_{rms}	root-mean-square voltage (V)
L	inductance (H)
f	wave frequency (Hz)
λ	wavelength (m)

Sources

Concept-first coverage of the course plan. Chapters 23-31 have lecture support; inductance, AC and EM waves are course-plan extensions without separate lecture files in this folder. Advanced field integrals are recognition topics, not worked derivations.

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

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